

MediGuide AI

Interactive AI-Powered Healthcare Information &
Triage Assistant

Project Overview & Presentation Slide Deck

01. Problem Statement

Information Overload

Patients struggle to find clear, reliable, and trustworthy medical guidance online during initial symptom onset, often encountering conflicting advice.

Overburdened Systems

Non-emergency cases frequently clog emergency rooms and healthcare clinics due to a severe lack of accessible preliminary triage tools.

Accessibility & Latency

Obtaining immediate professional advice for basic health questions is often slow, expensive, or completely inaccessible in remote settings.

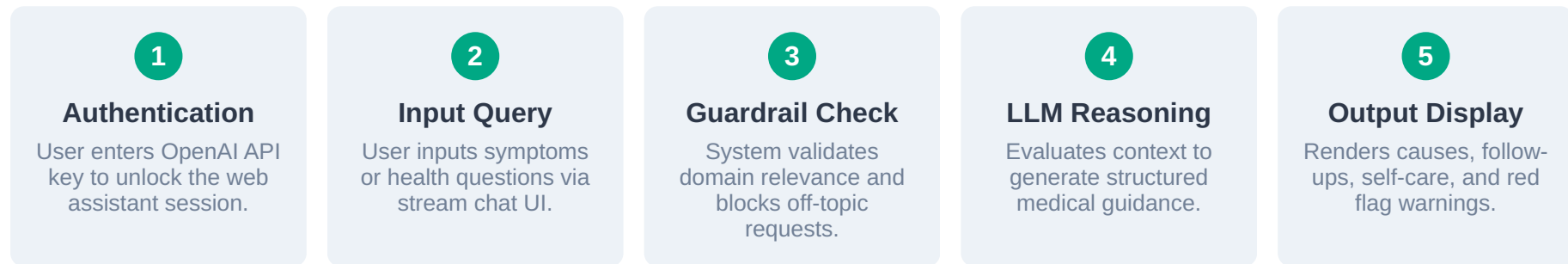
02. The Solution – MediGuide AI

- **AI-Driven Health Portal:** An interactive, responsive web application designed for rapid symptom analysis, general health inquiries, and structured triage guidance.
- **Secure Access Control:** Streamlit-powered access portal ensuring authorized user sessions via personal OpenAI API keys.
- **Strict Domain Guardrails:** Purpose-built system prompts ensuring responses are constrained exclusively to health and medical topics, eliminating off-topic hallucinations.

Key Value Proposition: Bridges the gap between initial health inquiries and professional care through an intuitive, privacy-conscious, and safe AI assistant.

03. System Architecture & Workflow

The system follows a 5-stage secure processing pipeline:



04. Feature 1: Secure Access Portal

- **User-Driven Authentication:** Clean login interface requiring an active OpenAI API key before providing access to the medical assistant.
- **Real-time API Validation:** Instant background check ("Validating API key...") ensuring valid credentials and smooth API usage.
- **Privacy-First Design:** Ensures individual session tracking and key security without relying on exposed central API keys.

05. Feature 2: Intelligent Symptom & Triage Breakdown

MediGuide AI organizes patient communication into actionable, structured categories:

Potential Causes

Identifies potential underlying factors such as viral/bacterial infections, inflammatory conditions, or environmental factors.

Follow-Up Questions

Dynamically asks clarifying questions (e.g., severity level, duration, accompanying symptoms) to refine context.

Self-Care Measures

Provides practical, safe guidance on hydration, rest, OTC medication guidelines, and supportive comfort care.

06. Feature 3: Red Flags & Patient Safety

Emergency Red Flag Identification

Automatically highlights critical warning signs that necessitate immediate emergency medical intervention:

- High-grade fever ($> 103^{\circ}\text{F}$ / 39.4°C)
- Severe headache, stiff neck, or rash
- Difficulty breathing, chest tightness, or severe pain
- Persistent vomiting, confusion, or difficulty waking up

Mandatory Medical Disclaimer: Prominently displayed on every view to explicitly state educational purpose and encourage consultation with licensed healthcare providers.

07. Feature 4: Domain Guardrails & Safety

- **Strict Domain Restriction:** The AI assistant is specialized exclusively in health and medical topics.
- **Off-Topic Query Handling:** Automatically detects non-health queries (e.g., asking general knowledge questions) and responds with polite, safety-focused refusal prompts.
- **Consistency & Focus:** Keeps user interactions aligned with clinical safety guidelines and prevents ungrounded AI behavior.

08. Technical Stack & Implementation

Frontend Framework

Streamlit (Python)

Delivers a clean, responsive, real-time chat interface and modal key validation portal.

AI Engine

OpenAI GPT Models

Engineered with custom system prompts for structured markdown output and medical safety context.

State Management

Python / Session State

Dynamic state tracking for API key authorization, context memory, and prompt guardrails.

09. Conclusion & Future Roadmap

Summary: MediGuide AI empowers users with accessible, structured, and safe AI triage assistance while maintaining essential safety disclaimers and guardrails.

Future Enhancements:

- **Multilingual Support:** Expanding accessibility across diverse global languages.
- **Voice Assistant Integration:** Enabling hands-free voice input and speech interaction.
- **EHR & Provider Integration:** Seamlessly connecting triage data with primary care scheduling and emergency services directories.